Chapter 1

Factor Rings

1.1 Properties of Factor Rings

We saw in the previous chapter that $\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/\langle 6\rangle$ is isomorphic to $\mathbb{Z}_6$. It follows that $\mathbb{Z}/\langle 6\rangle$ is a ring. This is in general true for all factor rings, however we have yet to prove this. Once we show that they are rings, we can then explore their structure using isomorphisms and homomorphisms. First we need to define the ring operations of factor rings which requires the following theorem.

Recall definition **??***:* If I is an ideal in the ring R, then the set of all cosets of I is called the *factor ring* R modulo I and is denoted R/I.

- $R/I = \{r + I \colon r \in R\}$

R/I is called a factor ring because (no surprise) -it is a ring under addition and multiplication of modulo classes. Before we can show that it is a ring, or why I needs to be an ideal; we define the operations and provide some examples.

- The sum of $a+I$ and $b+I$ is $(a+b)+I$.

  $$a + I \oplus b + I = (a+b) + I$$

  * Don't forget that $x \in a + I \Rightarrow x = a + ri$ for some $i \in I$. The addition sign can be confusing here because, $a + I$ denotes a set -not an operation. Some texts write $a + I \oplus b + I$ as "simply"$a + I + b + I$.

- The product of $a+I$ and $b+I$ is $ab+I$

  $$a + I \otimes b + I = ab + I \text{ or } (a+I)(b+I) = ab + I$$

The symbols $\oplus$ and $\otimes$ are used to distinguish the factor ring operation from the coset notation. Though often used it is by no means standard. So be careful when reading different texts! The theorem below is just theorem **??** restated for ideals.

Theorem 1. *Let I be an ideal in a ring R. If $a+I = b+I$ and $c+I = d+I$ in R/I, then*

$$(a+c) + I = (b+d) + I \textit{ and } ac + I = bd + I$$

Exercise 1. ★ Prove the theorem above.

Example 1. $\langle 5\rangle$ is a principal ideal in $\mathbb{Z}$. Addition and multiplication of cosets is just the same as addition and multiplication of congruence classes $\mathbb{Z}$ modulo.

$\mathbb{Z}/\langle 5\rangle = \{0+5\mathbb{Z}, 1+5\mathbb{Z}, 2+5\mathbb{Z}, 3+5\mathbb{Z}, 4+5\mathbb{Z}\}$. We can add and multiply as follows:

$$(2+5\mathbb{Z}) \oplus (1+5\mathbb{Z}) = 3+5\mathbb{Z}$$

and

$$(3+5\mathbb{Z}) \otimes (2+5\mathbb{Z}) = 6+5\mathbb{Z} = (5)+(1+5\mathbb{Z}) = (0+5\mathbb{Z}) \oplus (1+5\mathbb{Z}) = 1+5\mathbb{Z}$$

$$\begin{aligned} a+I \oplus b+I &= (a+b)+I && \text{closed under addition} \\ a+I \oplus (b+I \oplus c+I) &= (a+b+c)+I && \text{additively associative} \\ a+I \oplus 0+I &= a+I && \text{additive identity} \\ a+I \oplus -a+I &= 0+I && \text{additive inverse} \end{aligned}$$

$$\begin{aligned} a+I \otimes b+I &= ab+I && \text{closed under multiplication} \\ a+I \otimes (b+I \otimes c+I) &= abc+I && \text{multiplicatively associative} \\ a+I \otimes (b+I \oplus c+I) &= a+I \otimes ((b+c)+I) && \text{distribution} \\ &= (a(b+c))+I = (ab+ac)+I \end{aligned}$$

So R/I is a ring.

Thus $\mathbb{Z}/\langle 5\rangle = \mathbb{Z}/5\mathbb{Z}$ is a ring. To see that these are the distinct cosets, consider $a+5\mathbb{Z}$ for some integer $a > 5$. $a = 5q+r$ by the division algorithm so $a \equiv r \bmod 5$ for some integer $0 \le r < 5$. Thus a is an element in one of the above cosets, and it follows that $a+5\mathbb{Z} = r+5\mathbb{Z}$. As the operations $\mathbb{Z}/\langle 5\rangle$ are just modulo 5 arithmetic, we can also see that $\mathbb{Z}/\langle 5\rangle = \mathbb{Z}_5$.

Exercise 2. Give two examples of elements in $\mathbb{Z}/4\mathbb{Z} = \mathbb{Z}/\langle 4\rangle$, and add and multiply them.

Exercise 3. Find all the distinct cosets of $\mathbb{Z}/4\mathbb{Z}$.

Example 2. $3\mathbb{Z}/9\mathbb{Z} = \{0+9\mathbb{Z}, 3+9\mathbb{Z}, 6+9\mathbb{Z}\}$. For example $300 = 33 \cdot 9 + 3$ so $300 \equiv 3 \bmod 9$ and $300 \in 0+9\mathbb{Z}$.

If F is a field and $p(x)$ is a polynomial in $F[x]$, and I is the principal ideal $\langle p(x)\rangle$, then the cosets of I are the congruence classes modulo $p(x)$. Thus addition and multiplication of cosets is done exactly as in chapter **??**, and we have that $F[x]/I$ is the congruence class ring $F[x]/\langle p(x)\rangle$.

Example 3. Let D be the ring of rational numbers (in reduced terms) with odd denominators, and let E be the set of elements in D with even numerators. As we saw in exercise **??**, D/E has exactly two distinct ideals. $0+E$ and $x+E$ where x is a rational number with an odd denominator and odd numerator. Since $x-1 = \frac{2n+1}{2m+1} - \frac{2m+1}{2m+1} = \frac{2(n-m)}{2m+1} \in I$. So then $x+I = 1+I$. So the two distinct cosets are $0+E$ and $1+E$. By the theorem 1,

$\oplus$	$0+E$	$1+E$
$0+E$	$0+E$	$1+E$
$1+E$	$1+E$	$0+E$

and

$\otimes$	$0+E$	$1+E$
$0+E$	$0+E$	$0+E$
$1+E$	$0+E$	$1+E$

so we can see that D/E is also a ring.

Theorem 2. *Let I be an ideal in a ring R then,*

1. *R/I is a ring with addition and multiplication as defined above.*
2. *If R is abelian then R/I is also abelian.*
3. *If R has an identity then R/I has an identity.*

Proof. (1) First we need to show that R/I is an abelian group under addition. Recall from exercise **??** that $R/0 = R$. R is a ring so then $R/0$ is also a ring. We show that R/I is a subring of $R/0$. $R/I = \{r+I : r \in R\} \neq \emptyset$ since $I \neq \emptyset$ and $R \neq \emptyset$ as they are both rings. Now let $a+I, b+I \in R/I$. R is a ring; so $-b \in R$, $a-b \in R$, and $ab \in R$. Thus $a+I \oplus -b+I = (a-b)+I \in R/I$, and $a+I \otimes b+I = ab+I \in R/I$. So R/I is subring (and hence a ring) by the subring test.

Alternatively, we could show it is a ring directly as in example 1.

(2) Suppose that R is abelian so that $ab = ba$ for all $a, b \in R$. This implies that $a+I \otimes b+I = ab+I = ba+I = b+I \otimes a+I$.

(3) Suppose that R has an identity, say 1_R. This implies that $a+I \otimes 1_R+I = a1_R+I = a1_R+I = 1_R+I \otimes a+I$. So the identity of R/I is 1_R+I. □

- Part 2 is a sufficient but necessary condition for R/I to be commutative. For example R may not be non-abelian, but $R/R = 0_R$ is certainly abelian.

Theorem 3. *Let R be a ring. R/I is a ring if and only if I is an ideal of R.*

Proof. ($\Leftarrow$) From theorem 2 I is an ideal $\Rightarrow$ R/I is a ring.

($\Rightarrow$) Let R/I be a ring and I a subring of R, but I is not an ideal of R. Then there exists $a \in I$ such that $ra \notin I$ or $ar \notin I$ for some $r \in R$. Assume that $ra \notin I$. Since $a \in I$, $a+I = 0+I$ (cosets are equal or disjoint). Now consider $r+I$. $(r+I)(a+I) = ra+I$, but $(r+I)(a+I) = (r+I)(0+I) = 0+I = I$. So then $ra+I = 0+I$, a contradiction since $ra \notin I$. Similarly we get a contradiction if $ar \notin I$. So multiplication is not well defined and the set of cosets is not a ring.

Thus any ideal I in R gives us a factor ring by theorem 2, and any subring A which is not an ideal gives us a non-ring R/A. □

Exercise 4. Let $S = \{a+bi \colon b \in 2\mathbb{Z}\}$ (b is even and $i = \sqrt{-i}$). Prove that S is a subring of $\mathbb{Z}[i]$ (see exercise 4) but that S is not an ideal of $\mathbb{Z}[i]$.

Exercise 5. Let $S = \{a+bi \colon b \in 2\mathbb{Z}\}$ (b is even and $i = \sqrt{-i}$). Prove that $\mathbb{Z}[i]/S$ is not a ring.

Exercise 6. Prove that $\mathbb{R}[x]/\left\langle x^2+1\right\rangle$ is a ring. (We know that the ring of polynomials with integer coefficients, $\mathbb{R}[x]$, is a ring).

Example 4. Let $\langle 4-i\rangle \in \mathbb{Z}[i]$. $\mathbb{Z}[i] = \{x+yi \colon x, y \in \mathbb{Z}\}$ is called the *Gaussian integers*, and. $\langle 4-i\rangle = \{(4-i)\cdot z \colon z \in \mathbb{Z}[i]\}$. For any two Gaussian integers z and z' we have $(4-1i)z - (4-i)z' = (4-i)(z-z') \in \langle 4-i\rangle$, and $z_1(z_2(4-i)) = (z_2(4-i))z_1 = z_1z_2(4-i) \in \langle 4-i\rangle$. So $\langle 4-i\rangle$ is an ideal of $\mathbb{Z}[i]$.

By theorem3, $\mathbb{Z}[i]/\langle 4-i\rangle$ is a ring with elements (cosets) of the form $x+yi+\langle 4-i\rangle$ for $x, y \in \mathbb{Z}$. When dealing with cosets in $\mathbb{Z}[x]/\langle 4-i\rangle$ we can treat $4-i$ as 0 since $4-i+\langle 4-i\rangle = 0+\langle 4-i\rangle$. This allows us to simplify the set of *distinct* cosets. For example, since $4-i \equiv 0$ we also know that $4 \equiv i$ since $4-i+i = 4$. So then

$$6+3i+\langle 4-i\rangle = 6+12+\langle 4-i\rangle = 18+\langle 4-i\rangle$$

$4 \equiv i \Rightarrow 16 \equiv -1$ and $17 \equiv 0$. We can now further reduce the above coset to

$$18+\langle 4-i\rangle = 17+1+\langle 4-i\rangle = 1+\langle 4-i\rangle$$

As it turns out, there are 17 distinct cosets.

$$\mathbb{Z}[i]/\langle 4-i\rangle = \{0+\langle 4-i\rangle, 1+\langle 4-i\rangle, \ldots, 17+\langle 4-i\rangle\}$$

To see that this is the case, note that any other coset will be some integer multiple of $1+\langle 4-i\rangle$. Since $17(1+\langle 4-i\rangle) = 0+\langle 4-i\rangle$, $17(1+\langle 4-i\rangle)$ has order 1 or 17. If it is 1, then $1+\langle 4-i\rangle = 0+\langle 4-i\rangle$ and $1 \in \langle 4-i\rangle$ so $1 = (x+yi)(4-i) = 4x-ix+4yi+y = 4x+y+i(-x+4y)$ for integers x and y. Since $1 \in \mathbb{R}$ this implies that $1 = 4x+y$ and $0 = -x+4y$. Solving this system gives $y = 1/17 \notin \mathbb{Z}$, a contradiction. So the order of $\mathbb{Z}[x]/\langle 4-i\rangle$ is 17. From the operations we can also see that this ring is essentially the same as $\mathbb{Z}_{17}$, a field.

Exercise 7. Give two examples of elements in $\mathbb{Z}[i]/\langle 2-i\rangle$, and add and multiply them.

Exercise 8. Find all the distinct cosets of $\mathbb{Z}[i]/\langle 2-i\rangle$.

1.2 Prime and Maximal Ideals

Exercise 9. Let R be a commutative ring with unity and let $a_1, a_2, \ldots, a_n \in R$. Prove that

$$I = \langle a_1, a_2, \ldots, a_n\rangle = \{r_1a_1, r_2a_2, \ldots, r_na_n : r_i \in R\}$$

is an ideal in R. I is called the *ideal generated by* $a_1, a_2, \ldots, a_n$.

Exercise 10. If n is not prime prove that $\langle n\rangle$ is not a prime ideal in $\mathbb{Z}$

Exercise 11. Prove that the intersection of two prime ideals need not be prime.

Exercise 12. Prove that a nonzero integer p is prime if and only if the ideal $\langle p\rangle$ is maximal in $\mathbb{Z}$.

Exercise 13. Let F be a field and $p(x) \in F[x]$. Prove that $p(x)$ is irreducible if and only if the ideal $\langle p(x)\rangle$ is maximal in $F[x]$.

- All ideals of $\mathbb{Z}_n$ are principal.
- The ideals of $\mathbb{Z}_n$ are the divisors of n. To see this let $\langle m\rangle$ be an ideal in $\mathbb{Z}_n$. $0_R \in \langle m\rangle$ so then there is a $r \in R$ such that $rm = n \equiv 0 \bmod n$.
- The distinct divisors of n give distinct ideals of $\mathbb{Z}_n$. To see this let $0 < m < q$ be two distinct divisors of n. If $m|q$ then $\langle q\rangle \subset \langle m\rangle$. Otherwise $m \notin \langle q\rangle$.

Example 5. Consider $\mathbb{Z}_{36}$. The distinct ideals are $\langle 0\rangle$, $\langle 12\rangle$, $\langle 18\rangle$, $\langle 4\rangle$, $\langle 6\rangle$, $\langle 9\rangle$, $\langle 2\rangle$, $\langle 3\rangle$, and $\langle 1\rangle = \mathbb{Z}_{36}$.

- $\langle 0\rangle \subset \langle 18\rangle \subset \langle 9\rangle \subset \langle 3\rangle \subset \langle 1\rangle$
- $\langle 0\rangle \subset \langle 18\rangle \subset \langle 6\rangle \subset \langle 3\rangle \subset \langle 1\rangle$
- $\langle 0\rangle \subset \langle 18\rangle \subset \langle 6\rangle \subset \langle 2\rangle \subset \langle 1\rangle$

Continuing like this we can construct a lattice showing all the ideals and clearly see that $\langle 2\rangle$ and $\langle 3\rangle$ are the maximal ideals.

Exercise 14. Find all maximal ideals in $\mathbb{Z}_{10}$ and $\mathbb{Z}_{12}$.

Theorem 4. *Let R be a commutative ring with identity. R/I is an integral domain if and only if I is a prime ideal.*

Proof. ($\Rightarrow$) Suppose that R/I is an integral domain and $ab \in I$. Then $(a+I)(b+I) = ab + I = I = 0 + I$. So then ab equals the zero element of R. R is an integral domain so then $a = 0$ or $b = 0$. Thus $a \in I$ or $b \in I$ implying that I is prime.

($\Leftarrow$) Suppose that I is a prime ideal. R is commutative with identity, so we need only show that R has no zero divisors (nonzero elements a such that $a \cdot b = 0_{R/I}$ for any nonzero b). Suppose that $(a+I)(b+I) = ab + I = 0 + I = I$. So then $ab \in I$ and thus $a \in I$ or $b \in I$ since I is prime. So then $a+I$ OR $b+I$ is equal to $0+I$, and R/I is an integral domain. $\square$

Theorem 5. *Let R be a commutative ring with identity. R/I is a field if and only if I is a maximal ideal.*

Proof. ($\Rightarrow$) Let R/I be a field, and let M be an ideal of R such that $I \subset M$ (properly contains). We want to show that $M = R$. So there is a $m \in M$ such that $m \notin I$. So the $m + I \neq 0 + I$ and because R/I is a field there exists an $x \in R$ such that $(m + I)(x + I) = mx + I = 1 + I$.

Since $m \in M$ we have that $mx \in M$ (M is an ideal), and so $mx = 1 \in M$. By a previous exercise this implies that $M = R$. So I is maximal.

($\Leftarrow$) Let I be a maximal ideal. Let $m \in R$ but $m \notin I$. We want to show that $m + I$ must have a multiplicative inverse. Consider $M = \{mr + i \colon r \in R \text{ and } i \in I\}$. It is easy to see that M is an ideal in R, and $I \subset R$. I is maximal so then $M = R$. Thus $1 \in M$, say $1 = mx + i'$ where $i' \in I$. Then

$$1 + I = (mx + i') + I = (mx + I) \oplus (i' + I) = (mx + I) \oplus (0 + I) = mx + I$$

So then $mx + I = (m + I) \otimes (x + I)$. Thus $m + I$ has an inverse and R/I is a field. ☐

- A field is an integral domain, so theorem 5 tells us that a maximal ideal is always prime. Of course a prime ideal need not be maximal as the example below shows.

Example 6. $\langle x \rangle$ is a prime ideal in $\mathbb{Z}[x]$. $\langle x \rangle = \{f(x) \in \mathbb{Z}[x] \colon f(0) = 0\}$, zero constants. Thus if $g(x)h(x) \in \langle x \rangle$ then $g(0)h(0) = 0$ and either $g(0) = 0$ or $h(0) = 0$.

But $\langle x \rangle$ is not maximal since $\langle x \rangle \subset \langle x, 2 \rangle \subset \mathbb{Z}[x]$. It follows that $\mathbb{Z}[x]/\langle x \rangle$ is an integral domain but not a field.

1.3 Factor Rings and Homomorphisms

Factor rings arose as a generalization of congruence class arithmetic in the integers. Now we see that the concept of homomorphism and isomorphism that we studied in chapter **??** is closely related to factor rings and ideals.

- Recall definition **??**: a *(ring) homomorphism* is an order preserving function between rings, i.e., a homomorphism between rings R and S is a function, say f written $f \colon R \to S$, such that $f(ab) = f(a)f(b)$ and $f(a + b) = f(a) + f(b)$.

Theorem 6. *Let $f \colon R \to S$ be a ring homomorphism and let* $\ker f = \{r \in R \colon f(r) = 0_S\}$. *Then* $\ker f$ *is an ideal in R.*

Proof. f is a homomorphism so $f(0_R) = 0_S$ by theorem **??**, and we have that $\ker f \neq \emptyset$. Using the ideal test (theorem **??**), let $k, k' \in \ker f$. R is homomorphism so $f(k - k') = f(k) - f(k') = 0_S - 0_S = 0_S \Rightarrow r - r' \in \ker f$. Now for any$r \in R$ the homomorphic properties of f implies that $f(rk) = f(r)f(k) = f(r)0_S = 0_S \Rightarrow rk \in \ker f$. Similarly we can show that kr is in $\ker f$. So $\ker f$ is an ideal in R. ☐

- The ideal $\ker f$ from theorem 6 is called the *kernel of* f. It is simply the set of elements in R that is mapped to 0.

Example 7. Recall that the floor function (see example **??**) $f \colon \mathbb{R} \to \mathbb{Z}$ is a homomorphism. $\ker f = \{x \in \mathbb{Z} \colon 0 \leq x < 1\}$.

Exercise 15. Show that $\sigma \colon \mathbb{Z}[x] \to \mathbb{Z}$ defined by $\sigma(a_n x^n + a_{n-1}x^{n-1} \cdots + a_1 x + a_0) = a_0$ is a homomorphism and find $\ker f$.

Exercise 16. In exercise **??** we showed that $f \colon \mathbb{Z} \to \mathbb{Z}_n$ given by $k \to k \mod n$ is a ring homomorphism. Find $\ker f$.

Exercise 17. ★ Show that the $f \colon \mathbb{Z}_{12} \to \mathbb{Z}_4$ defined by $f(x) = x \bmod 4$ is a onto homomorphism, and find $\ker f$.

Exercise 18. Let $R = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b \in \mathbb{Z} \right\}$, and let $\phi\colon R \to \mathbb{Z}$ be defined by $\phi\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = a - b$. Show that ϕ is a homomorphism.

Exercise 19. Let ϕ be the function defined above in exercise 18. Find $\ker \phi$.

If $\ker f = R$, meaning all elements are mapped to 0, then f is the zero function, $f = 0$. If $\ker f$ has more than one element then $f(a) = f(b) = 0$ for some $a \neq b$ which is the definition of not being one-to-one (injective). So the kernel gives us a sense of how close a function is to being one-to-one which leads us to the following theorem.

Theorem 7. *Let $f\colon R \to S$ be a ring homomorphism with kernel $\ker f$. Then $\ker f = \{0_R\}$ if and only if f is one-to-one.*

Proof. ($\Leftarrow$) Suppose that f is one-to-one so that $f(a) = f(b) \iff a = b$ for any $a, b \in R$. Since f is a homomorphism $f(0_R) = 0_S$ by theorem **??**. Because f is one-to-one, $0_R, a \in \ker f \iff f(0_R) = f(a) = 0_S \iff 0_R = a$. Thus $\ker f = \{0_R\}$.

($\Rightarrow$) Suppose that $\ker f = \{0_R\}$ and $f(a) = f(b)$. We need to show that $a = b$. Because f is a homomorphism, $f(a - b) = f(a) - f(b) = 0_S \Rightarrow a - b \in \ker f$. By assumption $a - b = 0_R \Rightarrow a = b$. □

Exercise 20. Use theorem 7 to determine whether the functions from exercise 16 and 18 are one-to-one.

Theorem 6 tells us that every kernel is an ideal. Conversely, every ideal is a kernel of some homomorphism. This is where we connect factor rings and homomorphisms. Recall that if I is an ideal in R then R/I is also a ring. So we can define a function from the ring R to the ring R/I.

Theorem 8. *Let I be an ideal in the ring R. Then the function $\pi\colon R \to R/I$ defined by $\pi(r) = r + I$ is an onto homomorphism and $\ker \pi = I$.*

Proof. It is easy to see that π is onto since for any coset $r + I \in R/I$, $\pi(r) = r + I$ by definition. The homomorphism part follows by the definition of multiplication and addition in cosets. For any $r, r' \in R$

$$f(r + r') = (r + r') + I = (r + I) \oplus (r' + I)$$

and

$$f(r \cdot r') = (r \cdot r') + I = (r + I) \otimes (r' + I)\,.$$

The $\ker \pi$ will b e all the elements in $r \in R$ such that $\pi(r) = r + I = 0 + I = I$, and $r + I = I$ if and only if $r \in I$. Thus $\ker \pi = I$. □

- The function π from theorem 8 is called the *natural homomorphism*.

Exercise 21. Let R and S be rings. Show that $\pi\colon R \times S \to R$ defined by $\pi(r, s) = r$ is a onto homomorphism with $\ker f$ isomorphic to S.

The natural homomorphism is a special case of when $f\colon R \to S$ is an onto homomorphism. When this happens we say that f is *homomorphic image of* R. In the case that f is also one-to-one it is an isomorphism, and we know that R and S have identical structures. Meaning that the algebraic properties in one ring will be identical to those in the other ring. The next theorem relates the kernel of a homomorphic image and the rings.

Theorem 9. First Isomorphism Theorem. *Let $f\colon R \to S$ be a onto homomorphism of rings. Then the factor ring $R/\ker f$ is isomorphic to S.*

Proof. Let $\phi\colon R/\ker f \to S$. We want to show that ϕ is a isomorphism. To define ϕ we need to associate with each coset $r + \ker f \in R/\ker f$ an element of S. Let $\phi(r + \ker f) = f(r)$.

For ϕ to be well defined (i.e., a function) we need to show that if $r + \ker f = r' + \ker f$ then $f(r) = f(r')$.

- The problem is that a coset may possibly be represented by many different elements in R

If $r + \ker f = r' + \ker f$ then $r - r' \in \ker f$ by since $\ker f$ is an ideal, and since f is a homomorphism $f(r) - f(r') = f(r - r') = 0_S$. This $f(r) - f(r') = 0_S$ implies that $f(r) = f(r')$ as needed. Thus $\phi\colon R/\ker f \to S$ as defined by $\phi(r + \ker f) = f(r)$ is a function.

Now we need to show that ϕ is a isomorphism. If $s \in S$ then $s = f(r)$ for some $r \in R$ because f is onto. Then $s = f(r) = \phi(r + \ker f)$, and ϕ is also onto. If $\phi(r + \ker f) = \phi(r' + \ker f)$ then $f(r) = f(r') \Rightarrow f(r) - f(r') = 0_S \Rightarrow f(r - r') = 0_S$. Hence $r - r' \in \ker f$ which implies that $r + \ker f = r' + \ker f$. Therefore ϕ is also one-to-one.

Using the definition of addition and multiplication of cosets along with the homomorphic properties of f we show that ϕ is a homomorphism:

$$\begin{aligned} \phi\left[(r + \ker f) \oplus (r' + \ker f)\right] &= \phi\left[(r + r') + \ker f\right] = f(r + r') = f(r) + f(r') \\ &= \phi(r + \ker f) \oplus \phi(r' + \ker f) \end{aligned}$$

and

$$\begin{aligned} \phi\left[(r + \ker f) \otimes (r' + \ker f)\right] &= \phi\left[r \cdot r' + \ker f\right] = f(r \cdot r') = f(r)f(r') \\ &= \phi(r + \ker f) \otimes \phi(r' + \ker f) \end{aligned}$$

Thus ϕ is an isomorphism. □

$$\begin{tikzcd} R \arrow[r, "f"] \arrow[d, "\pi"'] & S \\ R/\ker f \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

Figure 1.1: A graph representing theorem 9.

This theorem states that every homomorphic image of a ring R is an isomorphism to $R/\ker f$ for some ideal $\ker f$. If you know all the factor rings of R then you also know all possible homomorphic images of R. The ideal $\ker f$ measures how much information is lost in the mapping from R to $R/\ker f$. When $\ker f = \{0_R\}$, no information is lost -it is an isomorphism.

- Theorem 9 implies that all homomorphic images of R can be determined using the factor rings of R.

The proof of theorem 9 shows that $\tilde{f} \circ \pi = f$, i.e., the graph in figure 1.1 is commutative.

Example 8. Recall that the principal ideal $\langle x \rangle$ in the ring $\mathbb{Z}[x]$ consists of all polynomials with constant term 0 (polynomial multiples of x). To understand $\mathbb{Z}[x]/\langle x \rangle$ we can look at the function $\sigma\colon \mathbb{Z}[x] \to \mathbb{Z}$ defined as in example 15 (it gives the constant term of a polynomial). σ is onto as for any integer $a \in \mathbb{Z}$, a is also a polynomial with integer coefficients so then $\sigma(a) = a$. In exercise 15 it was shown that σ is also a homomorphism. The kernel consists of all the elements in $\mathbb{Z}[x]$ mapped to 0. This is all polynomials without a constant term; which is just $\langle x\rangle$. So then $\ker\sigma = \langle x\rangle$. Thus by the first isomorphism theorem, $\mathbb{Z}[x]/\ker\sigma \cong \mathbb{Z}$[1].

Example 9. Let S be a homomorphic image of $\mathbb{Z}$. We want to classify all possible such images. If S is a homomorphic image of $\mathbb{Z}$ then there is a homomorphic onto function $f\colon \mathbb{Z} \to S$. If f is a homomorphism then the structure of S is exactly like that of the integers.

Now suppose that S is not isomorphic to $\mathbb{Z}$. Then the kernel of f is a nonzero ideal, i.e., $\ker f \neq \langle 0\rangle$. Because $\ker f$ is an ideal in $\mathbb{Z}$ it must be a principal ideal (a previous example). So then $\ker f = \langle n\rangle$ for some integer n. So then by theorem 9 $S \cong \mathbb{Z}/\langle n\rangle$ which we know is the same as $\mathbb{Z}_n$. So any homomorphic image of the integers is either $\mathbb{Z}$ or $\mathbb{Z}_n$ for some $n \in \mathbb{Z}$.

Exercise 22. Use theorem 9 to show that $R/\langle 0_R\rangle \cong R$.

Exercise 23. Use theorem 9 to show that $\mathbb{Z}_{20}/\langle 5\rangle \cong \mathbb{Z}_5$.

Exercise 24. Show that $R/\ker f \cong \mathbb{Z}$ where R and f are defined in exercise 18.

Exercise 25. Is the kernel $\ker f$ from 18 a prime ideal? Is it a maximal ideal?

[1]Recall that "$\cong$" notates "isomorphic to".